

Fundamentals of Forestry

Science

May, 2026

Fundamentals of Forestry (A07688)

Short Title: Fundamentals of Forestry
Faculty Science and Computing
Department Science
Cluster Forestry
Author ABOOTH
Credits 5

Level: Introductory

Description of Module / Aims

This module introduces the student to the world forestry resource and gives a brief overview of the history of forestry in Ireland. It provides students with an understanding of forest operations from year four (after establishment) through to harvesting, and onto the sawmill and timber end-uses. Along the way the student will be introduced to and learn some basic forestry skills such as pruning, brushing, thinning control and volume estimation.

Programmes

FORS-0018	BSc in Forestry (WD_SFORS_D)	1 / 2 / M
FORS-0018	BSc in Forestry (WD_SFORS_D)	2 / 4 / E

Indicative Content

- Biomes & forest biomes
- The world forest resource
- Irish forest history, natural history, native species, past management
- Forest operations: formative shaping & tending of broadleaves
- Forest operations: brashing, inspection paths and high pruning
- Forest operations: spacing and thinning, harvesting
- Forest products: log assortments, end-uses and the timber industry in Ireland
- Sustainable forest management and alternative silvicultural systems
- Alternative products from the forest
- Basic thinning volume estimation

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Explain forestry terminology, aspects of the world forest resource, and Irish forest history.
2. Discuss forest operations (brashing, thinning, pruning, harvesting) from year four to the end of the rotation.
3. Demonstrate and carry out key skills such as brashing, pruning, marking and volume estimation for thinning operations.
4. Operate and use basic tree measurement equipment used in forestry.
5. Distinguish the main forest products, and understand the alternative silvicultural systems that are available to forest managers.

Learning and Teaching Methods

- Lectures.
- Independent learning.
- Demonstrations.

- Field trips.

Assessment Methods

	Weighing	Outcomes Assessed
Final Written Examination	50%	1,2,5
Continuous Assessment	50%	
<i>Practical</i>	50%	3,4

Assessment Criteria

- <40%:** Very limited knowledge and understanding of the subject matter. Lacking in technical ability/skill and understanding to carry out assessments. Failure to meet the objectives of assessments. Unable to carry out or submit independent study and research. Unable to present written work with attention to correct formatting, grammar and spelling.
- 40%-49%:** Demonstrate a limited knowledge of the subject matter. Show some technical ability/skill in carrying out and reporting on assessments. Attempt made to meet objectives and carry out independent study and research. Final result lacks balance and arguments undeveloped. Can present written work correctly formatted with minor grammar and spelling errors only.
- 50%-59%:** Demonstrate satisfactory general knowledge of the main issues within the subject. Logically and competently carry out assessments. Satisfactory handling and awareness in carrying out independent study and research. Reporting and projects adequately structured and referenced.
- 60%-69%:** Demonstrate sound knowledge of the subject matter. Show ability to analyse and logically argue in an effective and mature style. Demonstrate initiative and the ability to criticise methods used and appreciation of experimental design and current knowledge limitations when carrying out assessments. Projects and reporting coherent and soundly structured illustrating wide and in depth reading on subject coupled with the proper use of references.
- 70%-100%:** Demonstrate authoritative handling of complex material and a highly developed and extensive understanding of the subject. Provide well focused analysis and convincing argument on the subject. Demonstrate imaginative approach, excellent technical ability, awareness, and aptitude in carrying out and reporting on assessments. Show initiative, individual thought, and enthusiasm in self study and independent learning.

Note:

- In addition to the normal regulations, a minimum of 35% must be achieved in both the final exam and continuous assessment components of the module. In cases where continuous assessment has a practical component attendance of at least 75% in the practicals is mandatory.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	24	
Field Trips	24	
Independent Learning	87	

Essential Material

- Forest Service, Anon. *_Code of best forest practice-Ireland_*. Ireland: Forest Service, 2000.
- Forest Service, Anon. *_Forestry Schemes Manual_*. Ireland: Forest Service, 2003.

- Hibberd, B.G. _Forestry Practice_. 11th Edition. UK: HMSO, 1991.
- Morgan, J.L. _Forest. Tree seedlings -best practice in supply, treatment and planting._ . UK: HMSO, 1999.
- Neeson, E.A. _A history of Irish forestry_. Ireland: Lilliput, 1991.